



Figure 2: NFW - changing the dynamics of the telecom world

Network function virtualisation

Network function virtualisation (NFV) is a new way to design, deploy, and manage networking services by decoupling the physical network equipment from the functions that run on them. It replaces hardware centric and dedicated network devices with software running on general-purpose CPUs or virtual machines, operating on standard servers.

NFV aims to transform the way network operators architect their networks, by evolving standard IT virtualisation technology to consolidate many network equipment types onto industry standard high volume servers, switches and storage, which could be located in data centres, network nodes and at end user premises, as illustrated in Figure 2.

These virtual appliances can be instantiated on demand without installing new equipment. For example, network operators may run an open source software-based firewall in a virtual machine (VM). This involves the implementation of network functions in software that can run on a range of industry standard servers and be moved to or instantiated in various locations in the network, as required. In other words, NFV promotes the implementation of network functions in software that can run on a range of standard IT hardware in data centres and be managed (moved, or replicated) without the need to modify the physical infrastructure.

The need for NFV

Traditional physical network hardware has always been difficult to change and upgrade. Introducing a software patch or rolling out a new service on a physical network can take months to complete and costs a lot.

NFV is a catalyst for structural market changes. It first started with mobile telecom networks, with the promise of making them more flexible and far easier to upgrade or change and to reduce hardware-related capex and opex.

As NFV developed further, service agility has become one of the main drivers in this space.

Compared with the current practices, NFV introduces the following major positive shifts:

- **Separation of software from hardware:** This enables the



Figure 3: Features of NFV

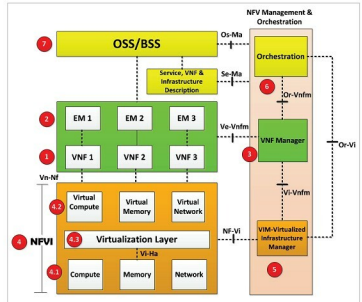


Figure 4: NFV architecture (Source: <http://www.telecomlighthouse.com/wp-content/uploads/2015/03/NFA-Architecture.png>)

software to evolve independently from the hardware, and vice versa.

- **Flexible deployment of network functions:** NFV can automatically deploy network-function software on a pool of hardware resources, which may run different functions at different times in different data centres.
- **Dynamic service provisioning:** Network operators can scale the NFV performance dynamically and on a grow-as-you-need basis with fine granularity, based on the current network conditions.

Salient features of NFV

- **Virtualisation:** Uses network resources without worrying about where the network is physically located, how big it is, how it is organised, etc.
- **Orchestration:** Manages thousands of devices.
- **Programmable:** Can change behaviour on-the-fly.
- **Dynamic scaling:** Can change size and quantity.
- **Automation**
- **Visibility:** Monitors resources and connectivity.
- **Performance:** Optimises network device utilisation.
- **Requires less space for network hardware.**
- **Reduced network power consumption and network maintenance costs.**
- **Easier network upgrades.**
- **Longer life cycles for network hardware.**

NFV architecture

1. **VNF (virtual network function):** A VNF is the basic block in NFV architecture. It is the virtualised network element.
2. **EM (element management):** EM is responsible for the functional management of VNF, i.e., fault, configuration, accounting, performance and security management (FCAPS). This may manage the VNFs through